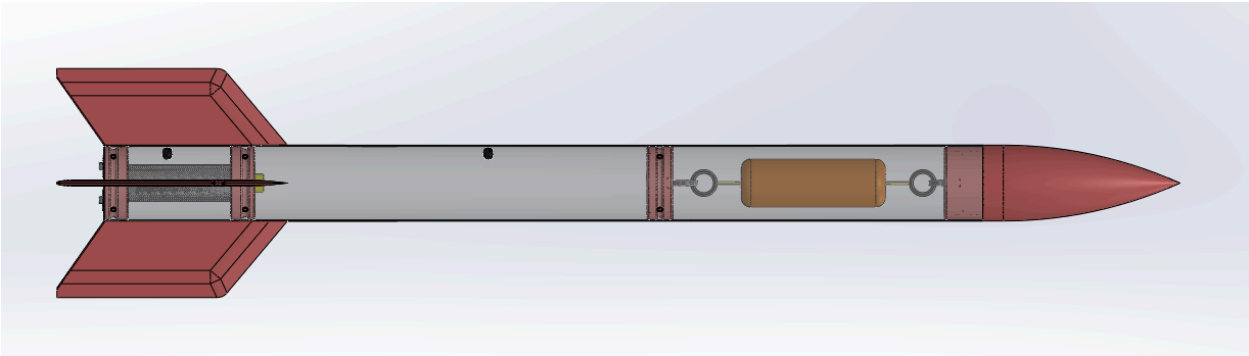
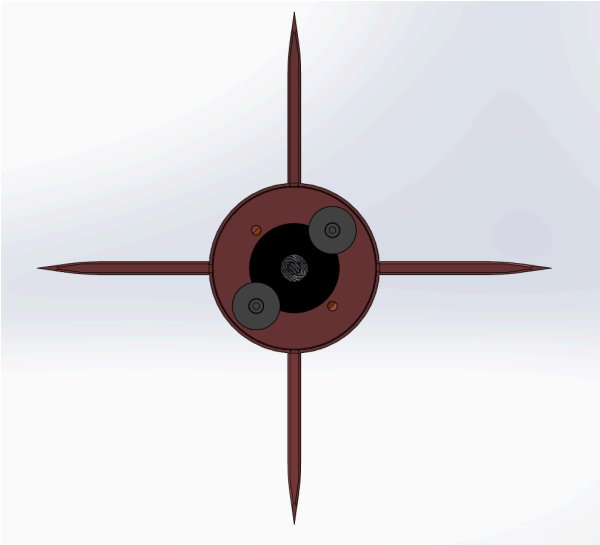
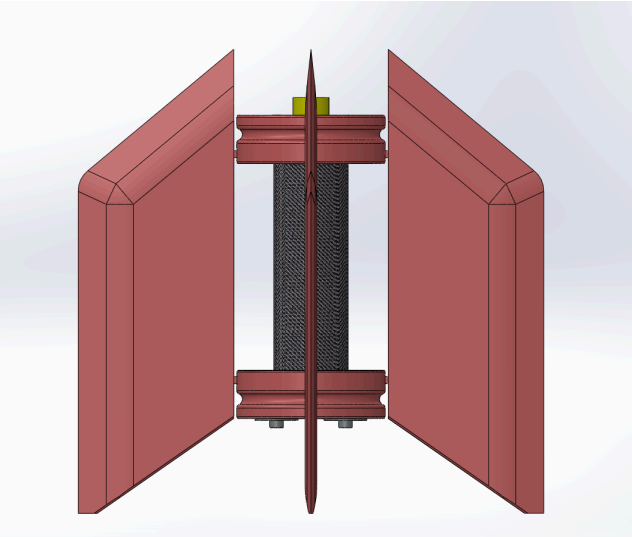
Ram Rocketry R1 Kit

Designed by Ian Leary

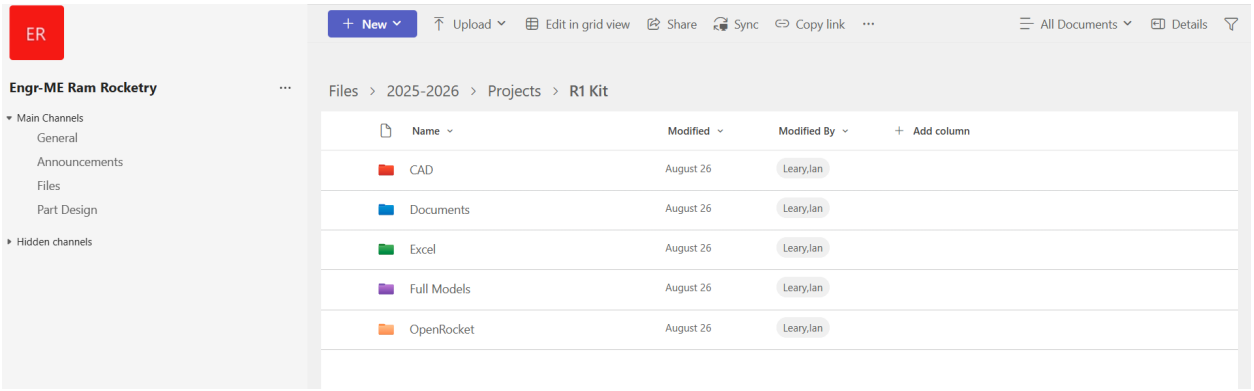








The Ram Rocketry R1 Kit is a simple, relatively easy to build rocket, allowing the builder to acquire their National Association of Rocketry (NAR) L1, and potentially L2 certifications. Designed for an Aerotech H100W-14A, but is capable of flying on I or J motors as well, as long as the motor is 38mm in diameter. All files required for this build are in the Ram Rocketry Teams Files channel, in the files tab, under 2025-2026 -> Projects -> R1, including all of the Solidworks files, stl files, OpenRocket simulation, and the materials Excel sheet.

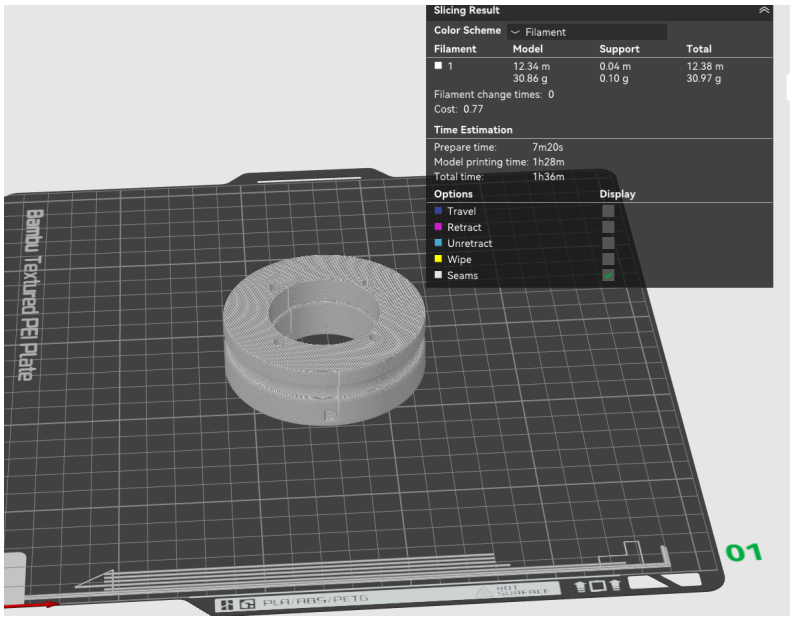


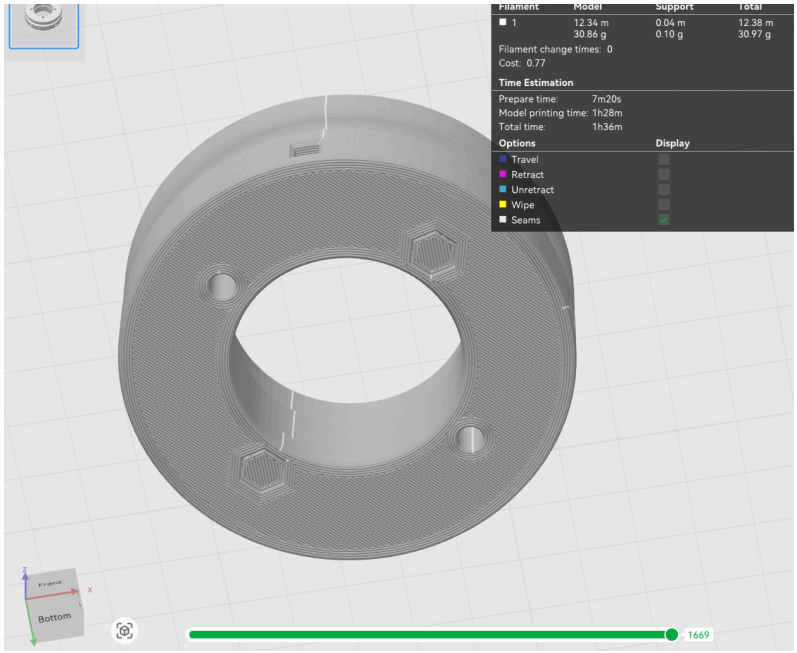
R1

Parts	Mass (Individual) (g)	Quantity	Mass (Combined) (g)	Fillament (g)	Price	Links
Motor Mount (Bottom)	33.350	1	33.350	33.150	$ -	
Motor Mount (Top)	32.180	1	32.180	32.180	$ -	
Motor (H100W-14A)	264.780	1	264.780	0.000	$ 54.99	
Airframe	264.510	1	264.510	0.000	$ 5.26	Airframe
Parachute Mount	44.340	1	44.340	44.340	$ -	
Parchute	93.480	1	93.480	0.000	$ 28.99	Parachute
Shock Cord	92.730	1	92.730	0.000	$ 9.99	Shock Cord
Fire Blanket	35.890	1	35.890	0.000	$ 8.95	Fire Blanket
Nose Cone	100.390	1	100.390	100.390	$ -	
Fins	56.130	4	224.520	225.800	$ -	
Square Pins	0.285	8	2.280	1.580	$ -	
1/4" x 2"Eyebolts	14.620	2	29.240	0.000	$ 3.99	Eyebolts
M4x25 Bolts	2.775	2	5.550	0.000	$ 1.99	Bolts
M4 Nuts	0.675	2	1.350	0.000	$ 0.50	Nuts
Fender Washers 5/32x7/8	5.490	2	10.980	0.000	$ 0.25	Fender Washers
Cork	3.500	1	3.500	0.000	$ 2.99	
Rail Buttons	3.000	2	6.000		$ 5.58	Rail Buttons
Quick Links	8.550	1	8.550	0.000	$ 7.99	Quick Links
NAR Membership					$ 30.00	
Totals			1253.620	437.440	$ 161.47	

Printing Components

Motor Mount Lower:



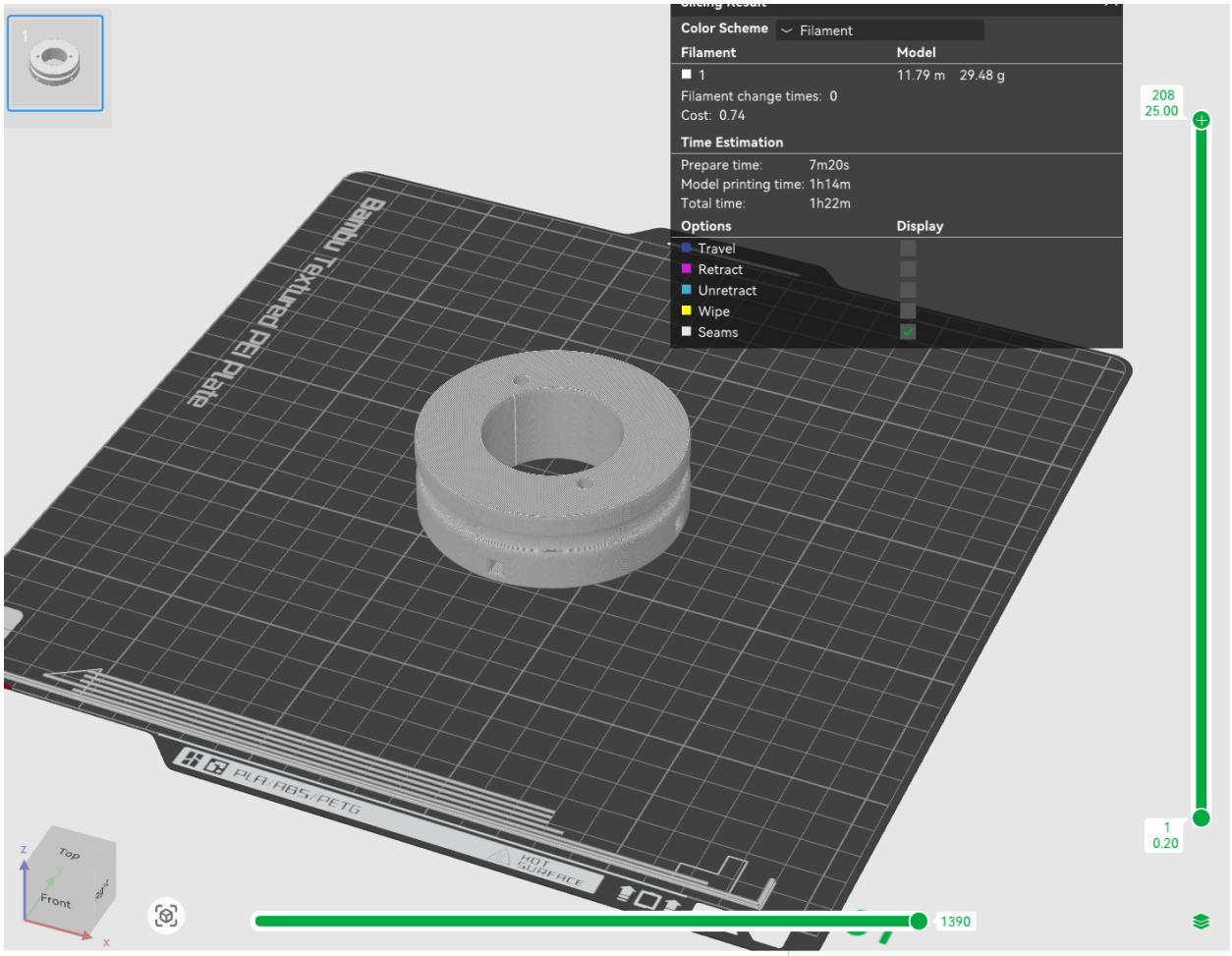


Infil: 15-20%

Walls: 4

Support: Only on hex insert sections touching build plate.

Motor Mount Upper:

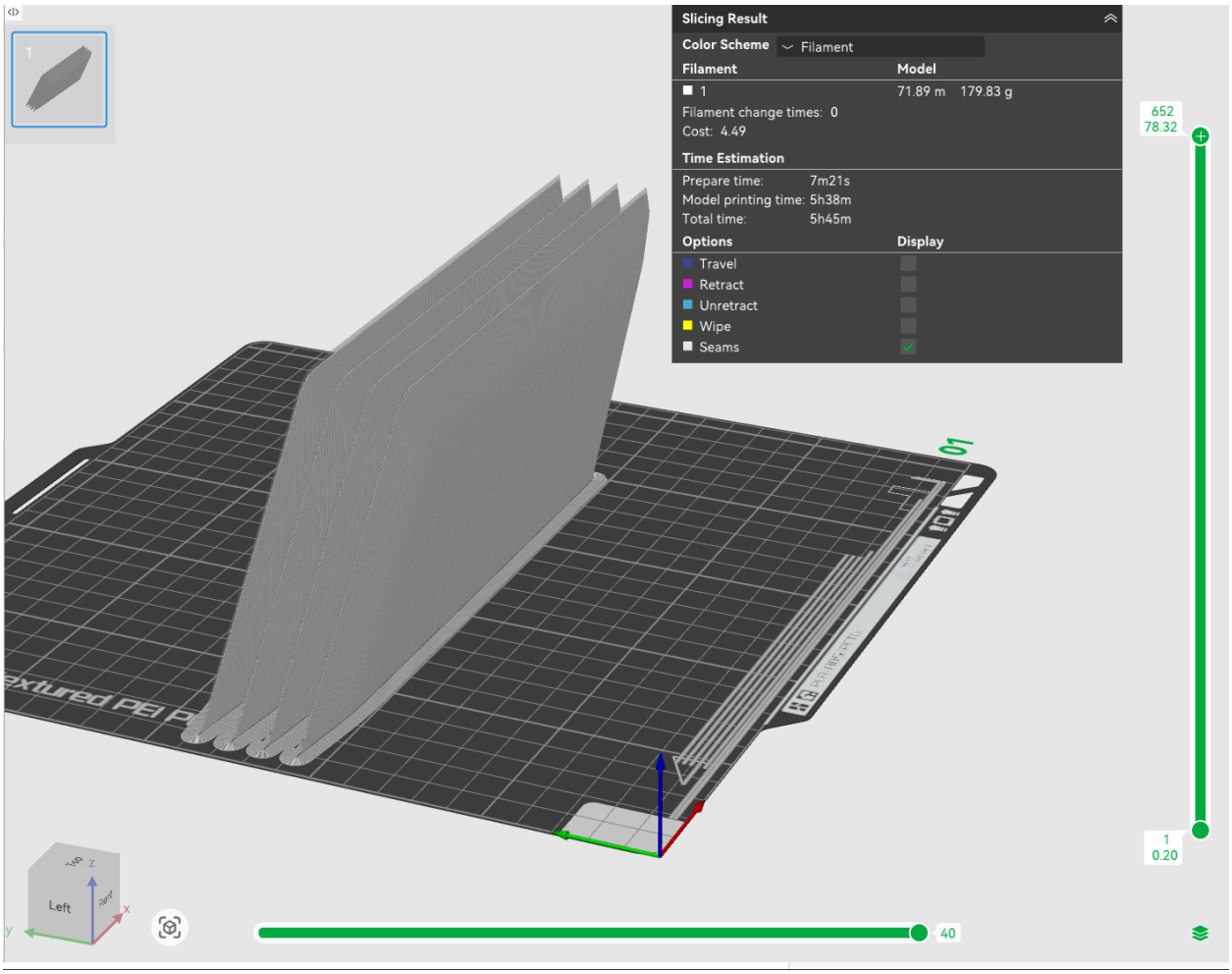


Infil: 15-20%

Walls: 4

Support: None

Fins:



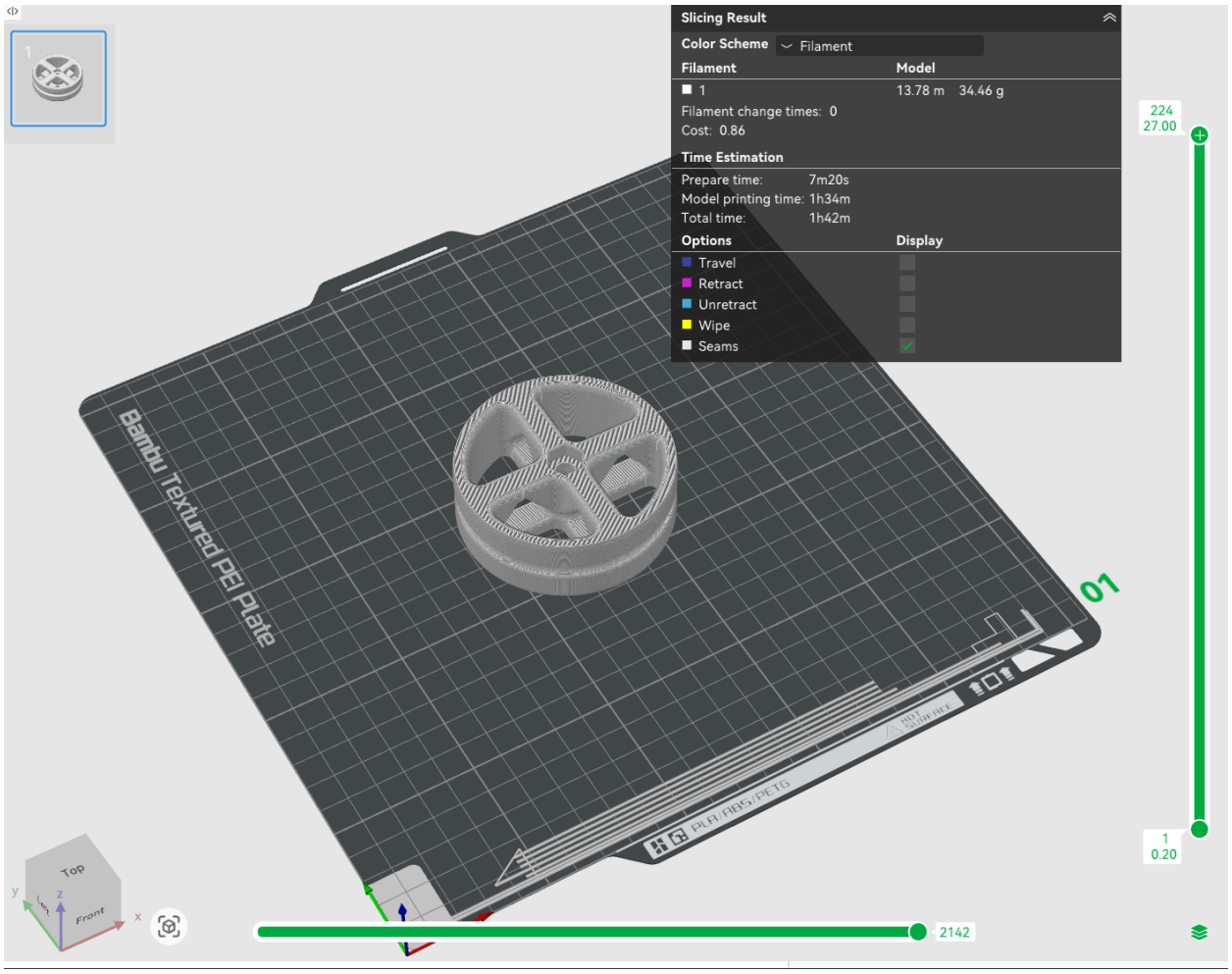
Infil: 20%

Walls: 4

Support: None

Outer Brim On

Parachute Mount:

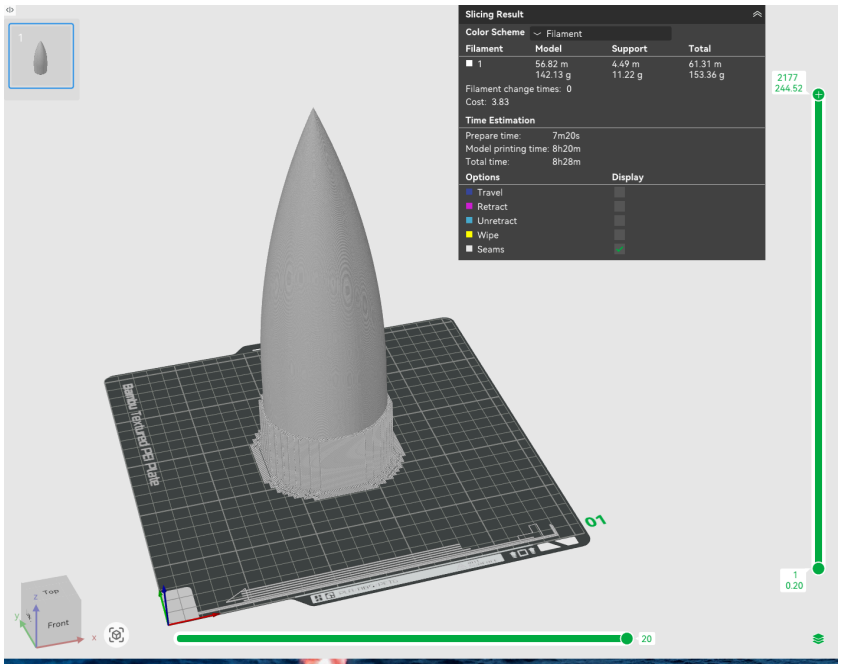


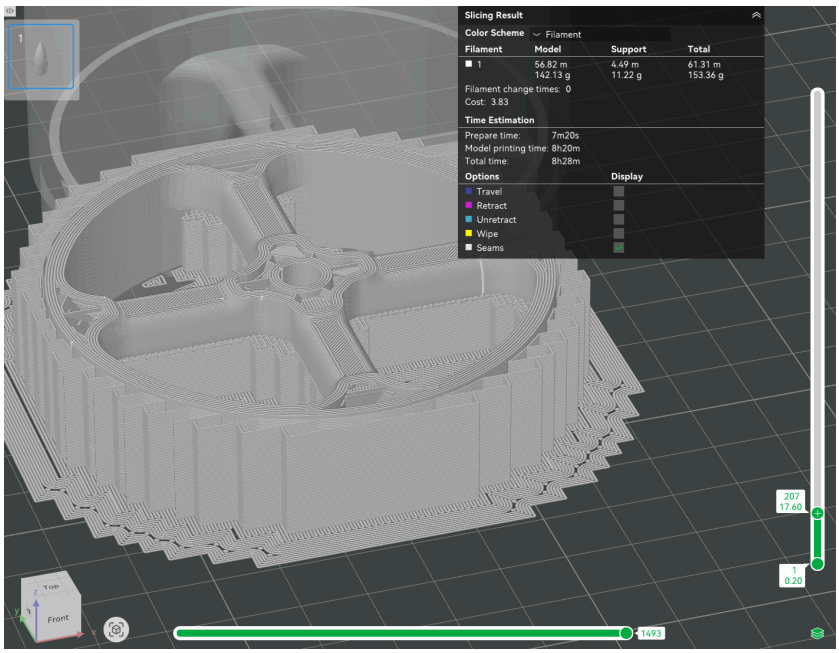
Infil: 20%

Walls: 4

Support: None

Nosecone:



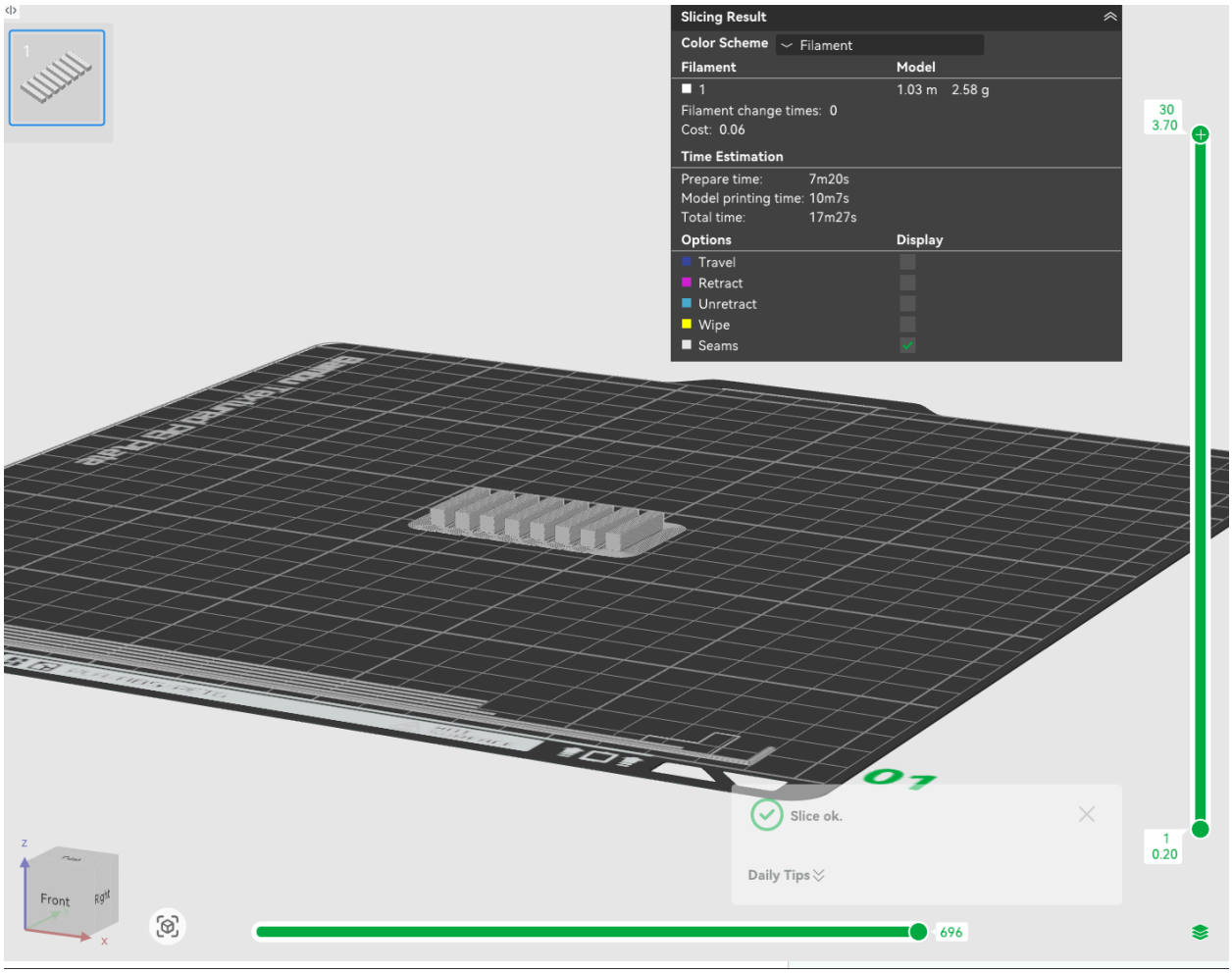


Infil: 15%

Walls: 4

Support: On, Touching build plate

Square Pins:



Infil: 85%

Walls: 4

Support: None

Outer Brim On

Assembly Procedure

1) Prepare Parts
    a) Before any epoxy is used, first make sure all parts fit correctly. 3D Printing is not always perfectly dimensionally accurate, especially using different types of materials.
    b) If parts fit together too tightly, especially the square pins to connect the fins to the motor mounts, lightly sand square pins (400-600 grit) until desired fit is achieved. Don't sand away too much material however, continuously check the fit as you go. Check the fit of the motor (if able) with the motor mounts as well.
    c) Check the fit of the fasteners, the M4 bolts/nuts, and the 1/4" Eyebolts/nuts. The Eyebolts will have to be threaded into the parachute mount/nosecone by design. **The M4 nuts will need to be superglued into place before epoxying the motor mount to the airframe.**
    d) If desired, the fins and nosecone can be sanded to achieve a smoother surface finish for a lower drag coefficient.
    e) Mark the location of the holes in the airframe using the hole template. It's crucial these holes are in the correct position. **Use tape to prevent the template from moving prior to marking the holes.**
    f) Drill holes in airframe. Start with a pilot hole, then work your way up to the desired diameter. Check the fit of the square pins, ensuring the pins can go through the holes without too much force. Once the holes are drilled,

sand/deburr any stray cardboard around the holes, on the inside and outside of the airframe.

2) Ensure all materials & equipment are present
    a) Epoxy can cure quickly (around 5 minutes if the ratio is correct), so make sure everything is accounted for **before starting the epoxy procedure.**
    b) Scale: For measuring out the two part epoxy. The final strength of the epoxy is extremely dependent on the ratio used, so **getting the ratio as close to stoichiometric is very important.**
    c) Disposable Syringes: Used for injecting the epoxy through holes in the airframe to designated sections of the components. At least two will be needed for the epoxy process.
    d) Disposable containers and stirrers for mixing the epoxy.

3) Assemble Components for Epoxy
    a) Insert the motor mounts into the airframe, oriented with the square holes in the mounts lined up with the holes in the airframe.
    b) **Insert rail buttons and tighten sufficiently**, as it is difficult to reach them after parts are epoxied.
    c) Insert the square pins into the motor mounts, and **test the fit of the fins onto the square pins.** If you have a spent motor, inserting it into the motor mounts can help keep the assembly square.

d) Once everything is lined up and the fins are parallel with the airframe, remove the fins from the assembly, but keep the square pins in the motor mounts. If used, remove the spent motor as well.
e) Put a small piece of tape covering 3 of the epoxy holes, leaving one uncovered to inject the epoxy in. This step is needed for the lower motor mount, upper motor mount, and parachute mount, so 9 holes covered by tape, 3 holes uncovered.

4) Epoxing Components
    a) Using the scale and disposable mixing container, **weigh out the two parts of the epoxy equally.**
    b) Mix thoroughly to ensure the epoxy ratio is uniform throughout.
    c) Use a disposable syringe to inject epoxy into the correct holes. It's recommended that you do two rounds of epoxy so it's not as rushed, the first set applied to the lower and upper motor mount, then the second round applied to the parachute mount.
    d) Inject until you feel pressure pushing back, ensuring the designated section is full. But occasionally **check the top and bottom of the parts looking down through the airframe to make sure epoxy isn't leaking out.**
    e) If able, insert the spent motor into the airframe to make sure the upper and lower mounts are still square.

f) Insert the fins onto the square pins for alignment before the epoxy is fully cured, but **make sure they won't be bonded yet** as they will be removed before their round of epoxy.

g) Repeat the process with the parachute mount. **Make sure the Eyebolt + nut are in place and tight before epoxing.**

h) Do not remove the tape covering the other holes for 6-8 hours

i) Once the internal components have fully cured in place, start a new round of epoxy.

j) Coat the surface of the square pins, and the face of the fin that will be flush with the airframe with epoxy.

k) Insert the fins onto the square pins, pressing firmly to ensure a strong bond. Rubber bands wrapped around the fins can be used to keep them pressed in place until the epoxy cures. Keep in mind **the epoxy is still liquid and can flow/drip away from where it was applied.**

l) Once the fins are bonded in place, add epoxy filets to the edges of the fins pressed against the airframe. To prevent the filets dripping, do this in several rounds of epoxy, letting one side cure before rotating to go to the next one. For highest quality filets, 4 rounds of epoxy is needed, but this can be done in 2 rounds with decent quality.

m) **Allow the epoxy to cure for 24 hours before any stresses are applied to the components.**

5) Finishing Touches

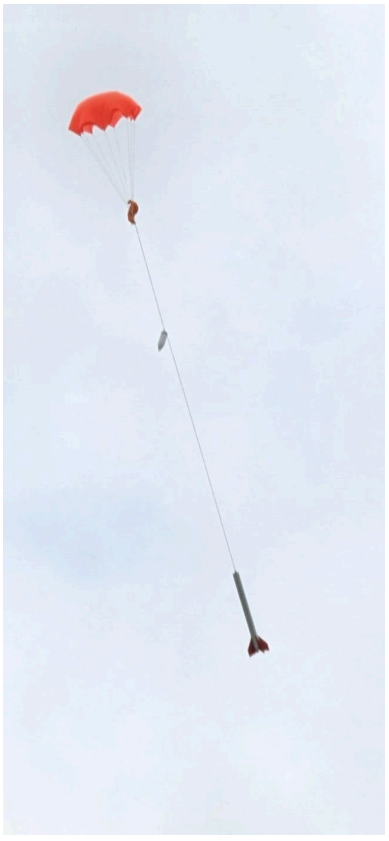
   a) Remove any tape still on the epoxied holes. Lightly sand rough areas on the airframe if needed.

   b) Check the fit of the motor in the mounts one last time to ensure there won't be any surprises at the launch site.

   c) Name and paint your rocket!

6) Parachute Packing

   a) Packing a parachute correctly is difficult and requires experience. However there will be a lot of people at the launch site that can help walk you through the process. Make sure you have a parachute (duh), fire blanket, shock cord, a cork, and two quick links.



7) L1 Certifications

a) Before the launch day, make sure you have an NAR Membership. You will need your membership number for the L1 paperwork, and there's not much cell service at the launch site so write it down somewhere.

b) The L1 Certification application can be found on NAR's website, https://www.nar.org/Level1HPRCertificationProcedures

Print it out prior to the launch day and fill out what you can (only the front side). The back side will be filled out by the NAR certifier.

c) Bring proof of an OpenRocket simulation of the rocket, including the stability margin. This can be a screenshot of the sim with the stability in the picture.

